

Lecture 3

Random Walks

Random walks are used to model stock price movements. Let S_0 be the initial value, and $S_1, S_2, \dots, S_t, \dots$ be the random values.

Basic Random Walk

The random walk is defined by:

$$S_t = z_1 + z_2 + \dots + z_t$$

where z_i are i.i.d.(independent and identically distributed) random variables. In particular,

$$\mathbb{E}[z_i] = \mathbb{E}[z_j]$$

and

$$\text{Var}(z_i) = \text{Var}(z_j).$$

Also, for $i \neq j$:

$$\text{Cov}(z_i, z_j) = 0, \quad \text{or equivalently, } \mathbb{E}[z_i z_j] = \mathbb{E}[z_i] \mathbb{E}[z_j].$$

Toy Example for Random Walk

Assume:

$$z_i = \begin{cases} 1 & \text{with probability } \frac{1}{2} \\ -1 & \text{with probability } \frac{1}{2} \end{cases}$$

Mean of S_t

If $S_t = z_1 + z_2 + \dots + z_t$, then:

$$\mathbb{E}[S_t] = \mathbb{E}[z_1] + \dots + \mathbb{E}[z_t] = t \mathbb{E}[z_i].$$

$$\mathbb{E}[z_i] = 1 \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$$

$$\mathbb{E}[S_t] = t \times 0 = 0$$

Variance of S_t

我们用方差的定义:

$$\text{Var}(S_t) = \mathbb{E} \left[(S_t - \mathbb{E}[S_t])^2 \right]$$

由于 $\mathbb{E}[S_t] = 0$, 所以:

$$\text{Var}(S_t) = \mathbb{E} \left[(z_1 + z_2 + \dots + z_t)^2 \right]$$

展开平方和:

$$(z_1 + z_2 + \cdots + z_t)^2 = \sum_{i=1}^t (z_i)^2 + \sum_{i \neq j} z_i z_j$$

所以:

$$\begin{aligned}\text{Var}(S_t) &= \sum_{i=1}^t \mathbb{E}[z_i^2] + \sum_{i \neq j} \mathbb{E}[z_i z_j] \\ \text{Var}(S_t) &= \mathbb{E} \left[(z_1 + \cdots + z_t - t \mathbb{E}[z_i])^2 \right]\end{aligned}$$

Expanding:

$$\text{Var}(S_t) = \mathbb{E} \left[\sum_{i=1}^t (z_i - \mathbb{E}[z_i])^2 + \sum_{i \neq j} (z_i - \mathbb{E}[z_i])(z_j - \mathbb{E}[z_j]) \right]$$

Since z_i and z_j are uncorrelated for $i \neq j$ (z_i and z_j 是独立的 (i.i.d.假设)):

$$\mathbb{E}[(z_i - \mathbb{E}[z_i])(z_j - \mathbb{E}[z_j])] = 0.$$

Thus:

$$\text{Var}(S_t) = \sum_{i=1}^t \text{Var}(z_i) = t \cdot \text{Var}(z_i).$$

最后再算出 $\text{Var}(z_i)$:

$$\text{Var}(z_i) = \mathbb{E}[z_i^2] - (\mathbb{E}[z_i])^2 = 1 - 0 = 1$$

所以:

$$\text{Var}(S_t) = t$$

Generalized Normal Random Walks—带漂移和波动的随机游走

The random walk is defined as:

$$S_t = r_1 + r_2 + \cdots + r_t$$

where

$$r_i = \mu + \sigma z_i,$$

with μ and σ being constants, and $z_i \sim \mathcal{N}(0, 1)$ (standard normal distribution). Thus, each r_i is normally distributed with mean μ and standard deviation σ :

$$r_i \sim \mathcal{N}(\mu, \sigma^2).$$

μ : 漂移参数 (drift) , 控制长期趋势 (如股票长期上涨的年化收益)

σ : 波动率 (volatility) , 控制短期波动幅度

We have:

$$\mathbb{E}[S_t] = t\mu, \quad \text{Var}(S_t) = t\sigma^2.$$

期望与方差的长期行为

$$\mathbb{E}[S_t] = \mathbb{E}[R_1] + \cdots + \mathbb{E}[R_t] = t\mu$$

长期来看，期望收益是线性随时间增长的。

因为每个 R_i 独立，方差可加：

$$\text{Var}(S_t) = \sum_{i=1}^t \text{Var}(R_i) = t\sigma^2$$

所以波动也是随时间平方根地增长。现实世界的股价常常用这个模型做初步拟合。

Autoregressive and Moving Average Random Walks

时间序列分析——自回归移动平均（ARMA）模型

自回归

当前的涨跌（收益）不仅受随机项影响，还受过去 p 期收益的线性组合影响。

Autoregressive: Reversion to Mean Property

$$S_t = r_1 + r_2 + \cdots + r_t$$

where:

$$r_i = c_0 + c_1 r_{i-1} + \cdots + c_p r_{i-p} + \sigma z_i.$$

Here:

- p is the number of lags(滞后阶数).
- If $p = 0$, we recover the normal random walk model.
没有滞后项时，就只剩下：

$$R_i = \mu + \sigma z_i$$

回到了上一页讲的广义随机游走。

Closer look at AR(1) Random Walk

We have:

$$r_i = c_0 + c_1 r_{i-1} + \sigma z_i$$

其中：

- c_0, c_1 是模型系数；

- σ 是噪声强度;
- $z_i \sim \mathcal{N}(0, 1)$ 标准正态白噪声;
- $p = 1$: 这是 AR(1) (一阶自回归) 。

Stationarity Condition 平稳性条件

- p is the number of lags.
- Typically, we require:

$$|c_1| < 1$$

to ensure stationarity.

平稳性条件: 防止过程发散、保证长期均值和方差存在。

- Assume:

$$\mathbb{E}[r_i] = \mathbb{E}[r_j], \quad \text{Var}(r_i) = \text{Var}(r_j), \quad \forall i, j.$$

- $z_i \sim \mathcal{N}(0, 1)$ (standard normal distribution).

Solving for Mean $\mathbb{E}[r_i]$

From(两边取期望):

$$\mathbb{E}[r_i] = c_0 + c_1 \mathbb{E}[r_{i-1}] + \sigma \mathbb{E}[z_i]$$

Since $\mathbb{E}[z_i] = 0$, we have:

$$\mathbb{E}[r_i](1 - c_1) = c_0$$

Thus:

$$\mathbb{E}[r_i] = \frac{c_0}{1 - c_1} \equiv \mu.$$

Rewrite the recursion

Subtract μ :

$$r_i - \mu = c_1(r_{i-1} - \mu) + \sigma z_i$$

Variance Recursion

$$\text{Var}(r_i) = \mathbb{E}[(r_i - \mu)^2]$$

Substitute:

$$= \mathbb{E} \left[(c_1(r_{i-1} - \mu) + \sigma z_i)^2 \right]$$

Since r_{i-1} and z_i are uncorrelated:

$$= c_1^2 \mathbb{E}[(r_{i-1} - \mu)^2] + \sigma^2 \mathbb{E}[z_i^2]$$

Thus:

$$\text{Var}(r_i) = c_1^2 \text{Var}(r_i) + \sigma^2$$

AR(1) Variance Solution

From:

$$\text{Var}(r_i)(1 - c_1^2) = \sigma^2$$

we have:

$$\text{Var}(r_i) = \frac{\sigma^2}{1 - c_1^2}.$$

Expectation and Variance of S_t

$$\mathbb{E}[S_t] = t\mu$$

$$\text{Var}(S_t) = \mathbb{E}[(S_t - \mathbb{E}[S_t])^2]$$

$$= \frac{\sigma^2}{1 - c_1^2} \left[t + 2 \sum_{i=1}^{t-1} (t - i)c_1^i \right].$$

(This is derived via algebraic summation expansion.)

Moving Average (MA) Random Walks

AR 模型: 当前值依赖于过去的值;

MA 模型: 当前值依赖于过去的随机冲击 (噪声) 。

Example:

$$r_i = \frac{1}{2} - \frac{1}{2}(r_{i-1} - r_{i-2}) + \sigma z_i$$

General Moving Average Model

Moving average model with q lags, $MA(q)$:

$$S_t = r_1 + r_2 + \cdots + r_t$$

where

$$r_i = \mu + (\phi_1 z_{i-1} + \cdots + \phi_q z_{i-q}) + \sigma z_i.$$

The past shocks influence r_i .

- q : 滞后阶数;
- $z_i \sim \mathcal{N}(0, 1)$: 标准正态白噪声;
- ϕ_j : 滞后权重, 控制过去噪声对当前的影响。

模型类型	当前收益依赖于	期望公式	方差公式
AR(1)	过去收益 R_{i-1}	$\mu = \frac{c_0}{1-c_1}$	$\frac{\sigma^2}{1-c_1^2}$
MA(1)	过去噪声 z_{i-1}	μ	$\phi_1^2 + \sigma^2$

Expectation

$$\mathbb{E}[r_i] = \mu$$

since all z_i are centered with zero mean.

Variance: MA(1) Example

For $MA(1)$:

$$r_i = \mu + \phi_1 z_{i-1} + \sigma z_i$$

Then,

$$\text{Var}(r_i) = \mathbb{E}[(r_i - \mu)^2] = \mathbb{E}[(\phi_1 z_{i-1} + \sigma z_i)^2].$$

Because z_{i-1} and z_i are uncorrelated:

$$= \mathbb{E}[(\phi_1 z_{i-1})^2] + \mathbb{E}[(\sigma z_i)^2] = \phi_1^2 + \sigma^2.$$

Variance of S_t for MA(1)

$$\mathbb{E}[S_t] = t\mu$$

$$\text{Var}(S_t) = \mathbb{E}[(S_t - \mathbb{E}[S_t])^2]$$

Expanding:

$$= \sum_{i=1}^t \mathbb{E}[(r_i - \mu)^2] + \sum_{i \neq j} \mathbb{E}[(r_i - \mu)(r_j - \mu)].$$

Since for $MA(1)$:

$$r_i = \mu + \phi_1 z_{i-1} + \sigma z_i,$$

we have that $r_i - \mu$ and $r_j - \mu$ are uncorrelated when $|i - j| \geq 2$.

Thus:

$$= t(\phi_1^2 + \sigma^2) + 2 \sum_{i=1}^{t-1} \mathbb{E}[(r_i - \mu)(r_{i+1} - \mu)]$$

We compute:

$$\mathbb{E}[(r_i - \mu)(r_{i+1} - \mu)] = \mathbb{E}[(\phi_1 z_{i-1} + \sigma z_i)(\phi_1 z_i + \sigma z_{i+1})] = \sigma \phi_1$$

Finally:

$$\text{Var}(S_t) = t(\phi_1^2 + \sigma^2) + 2(t-1)\sigma\phi_1$$

$$= (t-1)(\phi_1 + \sigma)^2 + \phi_1^2 + \sigma^2.$$

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ARMA(p, q) Model—自回归和移动平均合并在一起

The ARMA(p, q) process combines both AR(p) and MA(q) parts:

$$S_t = r_1 + r_2 + \cdots + r_t$$

where:

$$r_i = (c_0 + c_1 r_{i-1} + \cdots + c_p r_{i-p}) + (\phi_1 z_{i-1} + \cdots + \phi_q z_{i-q}) + \sigma z_i$$

with z_i representing standard noise.

其中每一个 R_i 满足:

$$r_i = (c_0 + c_1 r_{i-1} + c_2 r_{i-2} + \cdots + c_p r_{i-p}) \quad (\text{AR piece})$$

$$+ (\phi_1 z_{i-1} + \phi_2 z_{i-2} + \cdots + \phi_q z_{i-q}) \quad (\text{MA piece})$$

$$+ \sigma z_i \quad (\text{white noise piece})$$

- **AR部分**: 控制当前收益对过去收益的依赖;
- **MA部分**: 控制当前收益对过去噪声的依赖;
- **自噪声部分**: 始终存在的独立扰动项。

叠加后 $\rightarrow$ 可以捕捉大多数金融资产的收益率动态行为。

模型	定义公式	均值	方差	关键特征
Random Walk	$S_t = \sum z_i$	$t\mu$	$t\sigma^2$	独立增量, 价格随机上下波动
Generalized RW	$R_i = \mu + \sigma z_i$	μ	σ^2	有趋势漂移
AR(1)	$R_i = c_0 + c_1 R_{i-1} + \sigma z_i$	$\frac{c_0}{1-c_1}$	$\frac{\sigma^2}{1-c_1^2}$	均值回复, 递归性
MA(1)	$R_i = \mu + \phi z_{i-1} + \sigma z_i$	μ	$\phi^2 + \sigma^2$	短期平滑, 有记忆
ARMA(p, q)	综合 AR 和 MA	递归解	递归解	综合建模能力